Supplementary Methods and Artifact Record

SpecGuard-Chem 0.1.0

Daniel Hussey

Final university-feedback revision, 28 July 2026

1 Release status

This supplement accompanies the results manuscript *Before the Lab Acts: Benchmarking Language Models for Constrained Compound Selection*. The corrected data, deterministic baselines, and full 546-response provider matrix are complete, committed, and audited. Model comparisons use six complete 91-card pre-repair traces and six zero-call post-hoc-repaired views. No historical response or score from the earlier 50-card draft is reported here. Those artifacts are invalid because positional split references were previously misread as dataframe labels; they remain only as immutable provenance. Archival tagging and publication remain separate release-administration actions. Chemical-diversity and multi-round extensions are deferred to a separately versioned future analysis.

2 Artifact visibility contract

Field	Visibility	Role
Task, target, endpoint, source, activity semantics	System	Assay context and higher-is-better pChEMBL identity
Support IDs, structures, descriptors, pChEMBL activity	System	Project-local evidence observed before selection
Candidate IDs, structures, permitted descriptors	System	Finite action space
Constraints and budget	System	Executable action contract
Candidate activity	Evaluator only	Retrospective utility, ranking, and regret
Raw source paths and build-only metadata	Neither	Construction provenance, not model evidence

The public JSONL and scorer JSONL share benchmark, data, source, and transform provenance. Every scorer row carries the SHA256 digest of the canonical public card. The evaluator verifies identical task sets, candidate identity and order, provenance, and binding hashes before hydration. Non-oracle systems receive a strict allowlisted projection even when legacy monolithic fixtures are used.

3 Final data identity

Artifact	SHA256
Normalized corrected records	`cec651b6e97f044bf82820c465b441763cafa18a34b12361a33e79ab30faf438`
Build configuration	`425a40d3b64ac8398c49dfc04508e1c66522754c4e07e75e69810bc2517d9a6a`
System-input cards	`c18e66c726bb26f8afc3ba8422b21ec327444560d92750421f0dc44a2f393d9e`
Scorer outcomes	`96b5d6060e3c75dda34d835fd166fd074ca5621c18924aa0ea2714acba173ff4`
Card-build metadata	`d986ba96589032c59dac2dcc24cadf9aa3325616fa2a395cec682ece7220af54`
Card-build audit	`abb80ad110ef71d69a2cea2f09cc456399a81d431bc9e365320ab8b27064812c`
Exact 546-request export	`50e518893b19d4a7efd64c62e08ab94d610815f8fb7518c9af4b64ff40b6f6c5`
Pre-run cost estimate	`d11ce35da68e5082153be4bc57c027915ccb76cf8fb19b02e5adf767b2ab525d`

An independent final rebuild produced byte-identical system input, scorer outcomes, build metadata, and build audit. A second baseline run produced byte-identical per-system traces and score directories. A second request export produced the same 546 rows and request-file digest.

4 Task inclusion and exclusions

All 100 official `LO_All` task keys were considered. The inclusion rule keeps every task with at least ten candidates satisfying the declared candidate constraints. Ninety-one tasks are included. The following tasks were excluded without outcome-based selection:

CARA task key	Feasible candidates
CHEMBL1827734_IC50	0
CHEMBL3705332_IC50	0
CHEMBL3707531_IC50	0
CHEMBL3887078_IC50	0
CHEMBL3887456_EC50	0
CHEMBL3888518_Ki	0
CHEMBL3888871_Ki	0
CHEMBL3889212_IC50	6
CHEMBL3889266_IC50	8

The included endpoint distribution is 68 IC50, 17 Ki, 5 EC50, and one Potency task. Each included task has 50 support measurements and a distinct CARA target identifier. No ranking by candidate-pool size or other purposive subsampling was used. The feasibility rule is necessary for an exact top-10 action, but it limits inference to these 91 CARA lead-optimization cards; it does not establish representativeness of virtual-screening tasks, the excluded cards, or prospective projects.

5 Constraints

Output validity requires exactly ten unique candidate IDs, membership in the supplied candidate pool, and support-set exclusion. Candidate eligibility requires an RDKit-parseable structure, molecular weight no greater than 500 Da, and RDKit Crippen cLogP no greater than 4.5. The molecular-weight limit follows the conventional rule-of-five boundary. The prespecified cLogP cutoff is a simplified benchmark threshold, 0.5 below the conventional value of 5.0; it was fixed before model evaluation. It is not a universally validated medicinal-chemistry cutoff. The configured forbidden-SMARTS list is empty.

Table 2: Mutually exclusive candidate-constraint outcomes across all 19,588 official `LO_All` candidate records.

Outcome	Candidates	Percentage
Passed both thresholds	10,039	51.25%
Molecular weight $>$ 500 only	2,433	12.42%
cLogP $>$ 4.5 only	3,765	19.22%
Exceeded both	3,351	17.11%
Unparseable structure	0	0.00%

These rules are transparent eligibility controls for specification adherence, not claims of synthesis feasibility, oral bioavailability, selectivity, ADMET suitability, toxicity, or medicinal-chemistry approval.

6 Systems and frozen provider matrix

Deterministic systems comprise a random-valid selector, descriptor-desirability rules, similarity to the best observed support compound, per-task random-forest, gradient-boosting and linear support-vector regressors, and a non-deployable hidden-outcome oracle. Similarity and QSAR use 2,048-bit radius-2 Morgan fingerprints. Each regressor is fitted separately on one card's 50 visible support fingerprints and pChEMBL measurements. Random forest uses 100 trees, minimum leaf size 1, and seed 7. Gradient boosting uses the locked scikit-learn defaults and seed 7. Linear SVR uses a linear kernel, $C = 1.0$, and `StandardScaler(with_mean=False)`. There is no card-specific hyperparameter search. Predictions rank all feasible candidates in descending order; candidate ID breaks ties. Hidden candidate values are used only by the scorer and oracle.

The rules-only baseline and deterministic repair fallback use the same outcome-blind desirability score:

$$\begin{aligned} D(i) = 0.45 \max\left(0, 1 - \frac{|\mathrm{MW}_i - 350|}{350}\right) \\ + 0.35 \max\left(0, 1 - \frac{|\mathrm{cLogP}_i - 2.5|}{4.5}\right) \\ + 0.20 \max\left(0, 1 - \frac{|\mathrm{TPSA}_i - 75|}{150}\right). \end{aligned}$$

Feasible candidates are sorted by decreasing $D(i)$ with candidate ID as the tie-break. RDKit-computed TPSA enters this software fallback even for the basic LLM interface, which did not expose TPSA to the model. Random-valid shuffling seeds Python's random-number generator with the integer represented by the first eight hexadecimal digits of the SHA256 digest of seed 7, task ID, and `random_valid`, joined in that order by colons.

The provider matrix contains three conditions and two representations across 91 cards (546 requests):

- OpenAI `gpt-5.5-2026-04-23`, low reasoning effort and provider JSON-object response mode;
- Anthropic `claude-opus-4-8`, no extended thinking, using an explicit JSON-only prompt and deterministic one-object extraction; and
- DeepSeek `deepseek-v4-pro`, thinking disabled and provider JSON-object response mode, as checked on 16 July 2026, with the provider-returned model identifier retained in traces.

The basic interface supplies IDs, structures, molecular weight, and cLogP. The descriptor-enriched interface additionally supplies TPSA, hydrogen-bond donor and acceptor counts, and rotatable-bond count. Both receive the same action contract and an explicit instruction that pChEMBL is higher-is-better. The full ordered candidate pool is supplied on every card. Candidate outcomes are never supplied. Each condition uses a 4,096-token output limit; temperature and provider seed are unset. Provider SDK retries are disabled. Each recorded response therefore has one provider attempt, and malformed content remains part of pre-repair behavior rather than triggering a paid retry. The conservative uncached pre-run estimate is $106.06, with maximum estimated input 158,274 tokens and a hard cost ceiling of $120.

The staged pilot completed all six calls with one recorded attempt each, full provenance, and $0.449700535 aggregate usage-derived cost; cache-only replay reproduced every score artifact. The later authorized matrix completed all 546 responses. Each of the six raw conditions has one successful provider attempt per card and complete model, response, usage, latency, request-hash, and retained content evidence. The six repaired traces cover the same 91 cards and add zero provider calls. Usage-derived token-pricing cost is 58.957 dollars for the full matrix, with complete cost coverage.

The locked execution environment used Python 3.12.3, RDKit 2026.3.1, scikit-learn 1.8.0, NumPy 2.4.4, pandas 3.0.2, OpenAI SDK 2.36.0, and Anthropic SDK 0.100.0. The complete transitive environment and package hashes are recorded in `uv.lock`; exact model requests and cached responses are release artifacts.

7 Deterministic post-hoc repair

Repair is computed from the same raw response and adds zero provider calls:

1. Parse and validate the raw output against the action contract.
2. Retain valid raw candidate IDs in their original order.
3. Remove duplicates, out-of-pool IDs, support identities, and infeasible candidates.
4. Fill unoccupied slots using the deterministic rules-only ranking.
5. Revalidate and store both outputs and issue sets, the source-trace digest, the repair policy, and a zero-provider-call attribution field.

Raw metrics characterize model behavior. Repaired metrics characterize the model-plus-harness system and must never be relabeled as unaided model ability.

"Cards repaired" counts any contract modification. "Fallback positions" counts final candidate identities that were not retained from the valid, unique pre-repair sequence and were instead supplied by the fixed rules ranking. A card can therefore be repaired without receiving a fallback identity.

Table 3: Complete deterministic repair attribution. The per-repaired-card column reports mean; median (range) fallback positions.

Condition	Repaired	Identity changed	Fallback positions	Per repaired card	Zero fallback	All 10 fallback
OpenAI, basic	19/91 (20.88%)	10/91	14/910 (1.54%)	0.74; 1 (0–4)	9	0
OpenAI, descriptors	23/91 (25.27%)	10/91	15/910 (1.65%)	0.65; 0 (0–4)	13	0
Anthropic, basic	74/91 (81.32%)	71/91	252/910 (27.69%)	3.41; 2 (0–10)	3	1
Anthropic, descriptors	69/91 (75.82%)	68/91	229/910 (25.16%)	3.32; 3 (0–10)	1	1
DeepSeek, basic	83/91 (91.21%)	81/91	534/910 (58.68%)	6.43; 7 (0–10)	2	30
DeepSeek, descriptors	69/91 (75.82%)	69/91	355/910 (39.01%)	5.14; 4 (1–10)	0	12

8 Utility and statistical analysis

Let s_{mcr} be the ID returned by system m for card c at position r. Define a_{mcr} as $y_{s_{mcr},c}$ only when that position exists, the ID is a feasible candidate with an available hidden outcome, and the ID has not appeared at an earlier scored position; otherwise $a_{mcr} = 0$. Only the first k positions are scored, and feasible utility is

$$U_m(c) = \sum_{r=1}^{k} a_{mcr},$$

so only the first occurrence of an otherwise valid ID can score. Later duplicates, out-of-pool or infeasible IDs, unavailable outcomes, missing positions, and positions beyond k receive no credit. All labels have the common pChEMBL negative-log molar direction, but endpoint type and assay conditions differ. Absolute utility is therefore not interpreted as a biologically interchangeable cross-assay quantity.

System contrasts are paired within card. For systems A and B, the analysis first calculates $\Delta_c = U_A(c) - U_B(c)$ after both systems saw the same support set and candidate pool, then averages the 91 equally weighted differences. For valid fixed-size top-10 actions, a card-specific additive offset cancels from this contrast. NDCG@10 supplies a secondary within-card normalization against the ideal feasible ranking.

Marginal confidence intervals use 1,000 card-level bootstrap samples and NumPy's default random-number generator with seed 7. Each sample draws 91 cards with replacement, and the 2.5th and 97.5th percentiles of the sample means form the 95% interval. Headline contrasts use 2,000 paired bootstrap samples with seed 13: the 91 observed card-level differences are resampled with replacement, preserving pairing, and percentile limits are taken from the bootstrap mean differences. The reported proportion of bootstrap means above zero is a bootstrap replicate proportion, not a posterior probability. Intervals do not correct task selection, dependence beyond the card, repeated-call variation, or post-selection of the displayed best systems.